The Basics

Chapter One
Donella H. Meadows
Thinking in Systems: A Primer, 2009

I have yet to see any problem, however complicated,
which, when looked at in the right way,
did not become still more complicated.
—Poul Anderson[1]

More Than the Sum of Its Parts

A *system* isn't just any old collection of things.
A system is an interconnected set of elements
that is coherently organized in a way that achieves something.
If you look at that definition closely for a minute,
you can see that a system must consist of three kinds of things:
elements, interconnections, and a function or purpose.

For example, the elements of your digestive system
include teeth, enzymes, stomach, and intestines.
They are interrelated through the physical flow of food,
and through an elegant set of regulating chemical signals.
The function of this system is to break down food
into its basic nutrients
and to transfer those nutrients into the bloodstream
(another system),
while discarding unusable wastes.

A football team is a system
with elements such as players, coach, field, and ball.
Its interconnections are the rules of the game,
the coach's strategy,
the players' communications,
and the laws of physics
that govern the motions of ball and players.
The purpose of the team is to win games,
or have fun,
or get exercise,
or make millions of dollars,
or all of the above.

A school is a system.
So is a city,
and a factory,
and a corporation,
and a national economy.
An animal is a system.
A tree is a system,
and a forest is a larger system
that encompasses subsystems of trees and animals.
The earth is a system.
So is the solar system;
so is a galaxy.
Systems can be embedded in systems,
which are embedded in yet other systems.

Is there anything that is not a system?
Yes—a conglomeration without any particular
interconnections or function.
Sand scattered on a road by happenstance
is not, itself, a system.
You can add sand or take away sand
and you still have just sand on the road.
Arbitrarily add or take away football players,
or pieces of your digestive system,
and you quickly no longer have the same system.

When a living creature dies,
it loses its "*system-ness.*"
The multiple interrelations that held it together
no longer function,
and it dissipates,
although its material remains part
of a larger food-web system.
Some people say that an old city neighborhood
where people know each other
and communicate regularly
is a social system,
and that a new apartment block full of strangers is not—
not until new relationships arise
and a system forms.

You can see from these examples
that there is an integrity or wholeness about a system
and an active set of mechanisms to maintain that integrity.
Systems can change, adapt, respond to events,
seek goals, mend injuries,
and attend to their own survival in lifelike ways,
although they may contain or consist of nonliving things.
Systems can be self-organizing,
and often are self-repairing
over at least some range of disruptions.
They are resilient,
and many of them are evolutionary.
Out of one system
other completely new, never-before-imagined systems
can arise.

A system is more than the sum of its parts.
It may exhibit adaptive, dynamic, goal-seeking,
self-preserving, and sometimes evolutionary behavior.

Look Beyond the Players to the Rules of the Game

You think that because you understand "one"
that you must therefore understand "two"
because one and one make two.
But you forget that you must also understand "and."
—Sufi teaching story

The elements of a system
are often the easiest parts to notice,
because many of them are visible, tangible things.
The elements that make up a tree
are roots, trunk, branches, and leaves.
If you look more closely, you see specialized cells:
vessels carrying fluids up and down,
chloroplasts, and so on.
The system called a university
is made up of buildings, students, professors,
administrators, libraries, books, computers—
and I could go on and say
what all those things are made up of.
Elements do not have to be physical things.
Intangibles are also elements of a system.
In a university, school pride and academic prowess
are two intangibles
that can be very important elements of the system.
Once you start listing the elements of a system,
there is almost no end to the process.
You can divide elements into sub-elements
and then sub-sub-elements.
Pretty soon you lose sight of the system.
As the saying goes,
you can't see the forest for the trees.

Before going too far in that direction,
it's a good idea to stop dissecting out elements
and to start looking for the interconnections,
the relationships that hold the elements together.

Think About This

How to know whether you are looking at a system
or just a bunch of stuff:
– Can you identify parts? . . . and
– Do the parts affect each other? . . . and
– Do the parts together produce an effect
that is different from the effect of each part on its own?
. . . and perhaps
– Does the effect, the behavior over time,
persist in a variety of circumstances?

The interconnections in the tree system
are the physical flows and chemical reactions
that govern the tree's metabolic processes—
the signals that allow one part to respond
to what is happening in another part.
For example, as the leaves lose water on a sunny day,
a drop in pressure in the water-carrying vessels
allows the roots to take in more water.
Conversely, if the roots experience dry soil,
the loss of water pressure signals the leaves
to close their pores,
so as not to lose even more precious water.

As the days get shorter in the temperate zones,
a deciduous tree puts forth chemical messages
that cause nutrients to migrate out of the leaves
into the trunk and roots
and that weaken the stems,
allowing the leaves to fall.
There even seem to be messages
that cause some trees to make repellent chemicals
or harder cell walls
if just one part of the plant is attacked by insects.
No one understands all the relationships
that allow a tree to do what it does.
That lack of knowledge is not surprising.
It's easier to learn about a system's elements
than about its interconnections.

In the university system,
interconnections include the standards for admission,
the requirements for degrees,
the examinations and grades,
the budgets and money flows,
the gossip,
and most important,
the communication of knowledge
that is, presumably,
the purpose of the whole system.

Many of the interconnections in systems
operate through the flow of information.
Information holds systems together
and plays a great role in determining how they operate.

Some interconnections in systems
are actual physical flows,
such as the water in the tree's trunk
or the students progressing through a university.
Many interconnections are flows of information—
signals that go to decision points or action points
within a system.
These kinds of interconnections are often harder to see,
but the system reveals them to those who look.
Students may use informal information
about the probability of getting a good grade
to decide what courses to take.
A consumer decides what to buy
using information about his or her income,
savings, credit rating,
stock of goods at home, prices,
and availability of goods for purchase.
Governments need information
about kinds and quantities of water pollution
before they can create sensible regulations
to reduce that pollution.
(Note that information about the existence of a problem
may be necessary but not sufficient to trigger action—
information about resources, incentives, and consequences
is necessary too.)

If information-based relationships are hard to see,
functions or purposes are even harder.
A system's function or purpose
is not necessarily spoken, written, or expressed explicitly,
except through the operation of the system.
The best way to deduce the system's purpose
is to watch for a while
to see how the system behaves.

If a frog turns right and catches a fly,
and then turns left and catches a fly,
and then turns around backward and catches a fly,
the purpose of the frog has to do
not with turning left or right or backward
but with catching flies.
If a government proclaims its interest
in protecting the environment
but allocates little money or effort toward that goal,
environmental protection is not, in fact,
the government's purpose.
Purposes are deduced from behavior,
not from rhetoric or stated goals.

The function of a thermostat-furnace system
is to keep a building at a given temperature.
One function of a plant
is to bear seeds and create more plants.
One purpose of a national economy is,
judging from its behavior,
to keep growing larger.
An important function of almost every system
is to ensure its own perpetuation.

A Note on Language

The word function is generally used for a nonhuman system,
the word purpose for a human one,
but the distinction is not absolute,
since so many systems have both human and nonhuman elements.

System purposes need not be human purposes
and are not necessarily those intended
by any single actor within the system.
In fact, one of the most frustrating aspects of systems
is that the purposes of subunits
may add up to an overall behavior that no one wants.
No one intends to produce a society
with rampant drug addiction and crime,
but consider the combined purposes
and consequent actions of the actors involved:
– desperate people who want quick relief
from psychological pain
– farmers, dealers, and bankers who want to earn money
– pushers who are less bound by civil law
than are the police who oppose them
– governments that make harmful substances illegal
and use police power to interdict them
– wealthy people living in close proximity to poor people
– nonaddicts who are more interested
in protecting themselves
than in encouraging recovery of addicts

Altogether, these make up a system
from which it is extremely difficult
to eradicate drug addiction and crime.

Systems can be nested within systems.
Therefore, there can be purposes within purposes.
The purpose of a university
is to discover and preserve knowledge
and pass it on to new generations.
Within the university,
the purpose of a student may be to get good grades,
the purpose of a professor may be to get tenure,
the purpose of an administrator may be to balance the budget.
Any of those sub-purposes
could come into conflict with the overall purpose—
the student could cheat,
the professor could ignore the students in order to publish papers,
the administrator could balance the budget by firing professors.
Keeping sub-purposes and overall system purposes in harmony
is an essential function of successful systems.
I'll get back to this point later
when we come to hierarchies.

You can understand the relative importance
of a system's elements, interconnections, and purposes
by imagining them changed one by one.
Changing elements usually has the least effect on the system.
If you change all the players on a football team,
it is still recognizably a football team.
(It may play much better or much worse—
particular elements in a system can indeed be important.)
A tree changes its cells constantly,
its leaves every year or so,
but it is still essentially the same tree.
Your body replaces most of its cells every few weeks,
but it goes on being your body.
The university has a constant flow of students
and a slower flow of professors and administrators,
but it is still a university.
In fact it is still the same university,
distinct in subtle ways from others,
just as General Motors and the U.S. Congress
somehow maintain their identities
even though all their members change.
A system generally goes on being itself,
changing only slowly if at all,
even with complete substitutions of its elements—
as long as its interconnections and purposes remain intact.

If the interconnections change,
the system may be greatly altered.
It may even become unrecognizable,
even though the same players are on the team.
Change the rules from those of football to those of basketball,
and you've got, as they say, a whole new ball game.
If you change the interconnections in the tree—
say that instead of taking in carbon dioxide and emitting oxygen,
it does the reverse—
it would no longer be a tree.
(It would be an animal.)
If in a university the students graded the professors,
or if arguments were won by force instead of reason,
the place would need a different name.
It might be an interesting organization,
but it would not be a university.
Changing interconnections in a system
can change it dramatically.

The least obvious part of the system,
its function or purpose,
is often the most crucial determinant
of the system's behavior.

Changes in function or purpose also can be drastic.
What if you keep the players and the rules
but change the purpose—
from winning to losing, for example?
What if the function of a tree
were not to survive and reproduce
but to capture all the nutrients in the soil
and grow to unlimited size?
People have imagined many purposes for a university
besides disseminating knowledge—
making money, indoctrinating people, winning football games.
A change in purpose changes a system profoundly,
even if every element and interconnection remains the same.

To ask whether elements, interconnections, or purposes
are most important in a system
is to ask an unsystemic question.
All are essential.
All interact.
All have their roles.
But the least obvious part of the system,
its function or purpose,
is often the most crucial determinant of the system's behavior.
Interconnections are also critically important.
Changing relationships usually changes system behavior.
The elements,
the parts of systems we are most likely to notice,
are often (not always)
least important in defining the unique characteristics
of the system—
unless changing an element also results
in changing relationships or purpose.

Changing just one leader at the top—
from a Brezhnev to a Gorbachev,
or from a Carter to a Reagan—
may or may not turn an entire nation in a new direction,
though its land, factories,
and hundreds of millions of people
remain exactly the same.
A leader can make that land and those factories and people
play a different game with new rules,
or can direct the play toward a new purpose.

And conversely,
because land, factories, and people
are long-lived, slowly changing, physical elements of the system,
there is a limit to the rate
at which any leader can turn the direction of a nation.

Bathtubs 101—Understanding System Behavior over Time

Information contained in nature . . .
allows us a partial reconstruction of the past. . . .
The development of the meanders in a river,
the increasing complexity of the earth's crust . . .
are information-storing devices
in the same manner that genetic systems are. . . .
Storing information means increasing
the complexity of the mechanism.
—Ramon Margalef[2]

A *stock* is the foundation of any system.
Stocks are the elements of the system
that you can see, feel, count, or measure
at any given time.
A system stock is just what it sounds like:
a store, a quantity,
an accumulation of material or information
that has built up over time.
It may be the water in a bathtub,
a population,
the books in a bookstore,
the wood in a tree,
the money in a bank,
your own self-confidence.
A stock does not have to be physical.
Your reserve of good will toward others
or your supply of hope that the world can be better
are both stocks.

Stocks change over time
through the actions of a *flow*.
Flows are filling and draining,
births and deaths,
purchases and sales,
growth and decay,
deposits and withdrawals,
successes and failures.
A stock, then, is the present memory
of the history of changing flows within the system.

A stock is the memory of the history of changing flows
within the system.

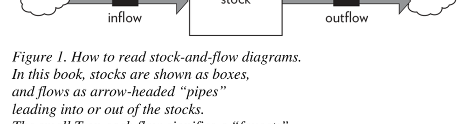

Figure 1. How to read stock-and-flow diagrams.
In this book, stocks are shown as boxes,
and flows as arrow-headed "pipes"
leading into or out of the stocks.
The small T on each flow signifies a "faucet;"
it can be turned higher or lower, on or off.
The "clouds" stand for wherever the flows come from and go to—
the sources and sinks that are being ignored
for the purposes of the present discussion.

For example, an underground mineral deposit is a stock,
out of which comes a flow of ore through mining.
The inflow of ore into a mineral deposit
is minute in any time period less than eons.
So I have chosen to draw (Figure 2)
a simplified picture of the system without any inflow.
All system diagrams and descriptions
are simplified versions of the real world.

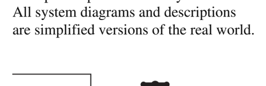

Figure 2. A stock of minerals depleted by mining.

Water in a reservoir behind a dam is a stock,
into which flow rain and river water,
and out of which flows evaporation from the reservoir's surface
as well as the water discharged through the dam.

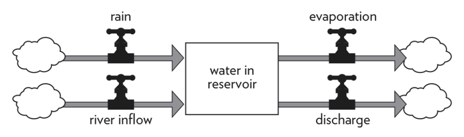

Figure 3. A stock of water in a reservoir
with multiple inflows and outflows.

The volume of wood in the living trees in a forest is a stock.
Its inflow is the growth of the trees.
Its outflows are the natural deaths of trees
and the harvest by loggers.
The logging harvest flows into another stock,
perhaps an inventory of lumber at a mill.
Wood flows out of the inventory stock
as lumber sold to customers.

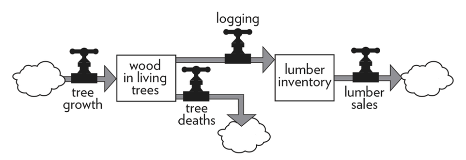

Figure 4. A stock of lumber linked to a stock of trees
in a forest.

If you understand the dynamics of stocks and flows—
their behavior over time—
you understand a good deal
about the behavior of complex systems.
And if you have had much experience with a bathtub,
you understand the dynamics of stocks and flows.

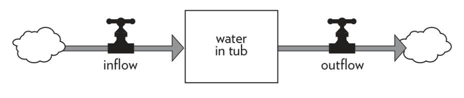

Figure 5. The structure of a bathtub system—
one stock with one inflow and one outflow.

Imagine a bathtub filled with water,
with its drain plugged up and its faucets turned off—
an unchanging, undynamic, boring system.
Now mentally pull the plug.
The water runs out, of course.
The level of water in the tub goes down
until the tub is empty.

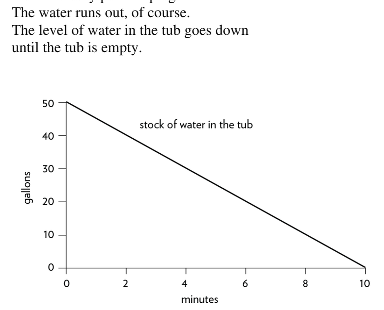

Figure 6. Water level in a tub when the plug is pulled.

A Note on Reading Graphs of Behavior over Time

Systems thinkers use graphs of system behavior
to understand trends over time,
rather than focusing attention on individual events.
We also use behavior-over-time graphs
to learn whether the system is approaching a goal or a limit,
and if so, how quickly.

The variable on the graph may be a stock or a flow.
The pattern—the shape of the variable line—is important,
as are the points at which that line changes shape or direction.
The precise numbers on the axes are often less important.

The horizontal axis of time
allows you to ask questions about what came before,
and what might happen next.
It can help you focus on the time horizon
appropriate to the question or problem you are investigating.

Now imagine starting again with a full tub,
and again open the drain,
but this time, when the tub is about half empty,
turn on the inflow faucet
so the rate of water flowing in
is just equal to that flowing out.
What happens?
The amount of water in the tub stays constant
at whatever level it had reached
when the inflow became equal to the outflow.
It is in a state of *dynamic equilibrium*—
its level does not change,
although water is continuously flowing through it.

Imagine turning the inflow on somewhat harder
while keeping the outflow constant.
The level of water in the tub slowly rises.
If you then turn the inflow faucet down again
to match the outflow exactly,
the water in the tub will stop rising.
Turn it down some more,
and the water level will fall slowly.

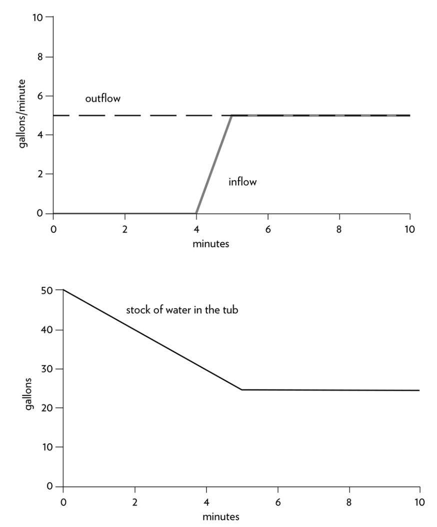

Figure 7. Constant outflow,
inflow turned on after 5 minutes,
and the resulting changes in the stock of water in the tub.

This model of a bathtub is a very simple system
with just one stock, one inflow, and one outflow.
Over the time period of interest (minutes),
I have assumed that evaporation from the tub is insignificant,
so I have not included that outflow.
All models, whether mental models or mathematical models,
are simplifications of the real world.
You know all the dynamic possibilities of this bathtub.
From it you can deduce several important principles
that extend to more complicated systems:
– As long as the sum of all inflows
exceeds the sum of all outflows,
the level of the stock will rise.
– As long as the sum of all outflows
exceeds the sum of all inflows,
the level of the stock will fall.
– If the sum of all outflows equals the sum of all inflows,
the stock level will not change;
it will be held in dynamic equilibrium
at whatever level it happened to be
when the two sets of flows became equal.

The human mind seems to focus more easily
on stocks than on flows.
On top of that, when we do focus on flows,
we tend to focus on inflows more easily than on outflows.
Therefore, we sometimes miss seeing
that we can fill a bathtub
not only by increasing the inflow rate,
but also by decreasing the outflow rate.
Everyone understands
that you can prolong the life of an oil-based economy
by discovering new oil deposits.
It seems to be harder to understand
that the same result can be achieved by burning less oil.
A breakthrough in energy efficiency is equivalent,
in its effect on the stock of available oil,
to the discovery of a new oil field—
although different people profit from it.

A stock can be increased by decreasing its outflow rate
as well as by increasing its inflow rate.
There's more than one way to fill a bathtub!

Similarly, a company can build up a larger workforce
by more hiring,
or it can do the same thing
by reducing the rates of quitting and firing.
These two strategies may have very different costs.
The wealth of a nation can be boosted by investment
to build up a larger stock of factories and machines.
It also can be boosted, often more cheaply,
by decreasing the rate at which factories and machines
wear out, break down, or are discarded.

You can adjust the drain or faucet of a bathtub—
the flows—abruptly,
but it is much more difficult to change the level of water—
the stock—quickly.
Water can't run out the drain instantly,
even if you open the drain all the way.
The tub can't fill up immediately,
even with the inflow faucet on full blast.
A stock takes time to change,
because flows take time to flow.
That's a vital point,
a key to understanding why systems behave as they do.
Stocks usually change slowly.
They can act as delays, lags, buffers, ballast,
and sources of momentum in a system.
Stocks, especially large ones,
respond to change, even sudden change,
only by gradual filling or emptying.

Stocks generally change slowly,
even when the flows into or out of them change suddenly.
Therefore, stocks act as delays or buffers
or shock absorbers in systems.

People often underestimate the inherent momentum of a stock.
It takes a long time for populations to grow or stop growing,
for wood to accumulate in a forest,
for a reservoir to fill up,
for a mine to be depleted.
An economy cannot build up a large stock
of functioning factories and highways
and electric plants overnight,
even if a lot of money is available.
Once an economy has a lot of oil-burning furnaces
and automobile engines,
it cannot change quickly
to furnaces and engines that burn a different fuel,
even if the price of oil suddenly changes.
It has taken decades to accumulate the stratospheric pollutants
that destroy the earth's ozone layer;
it will take decades for those pollutants to be removed.

Changes in stocks set the pace of the dynamics of systems.
Industrialization cannot proceed faster
than the rate at which factories and machines can be constructed
and the rate at which human beings can be educated
to run and maintain them.
Forests can't grow overnight.
Once contaminants have accumulated in groundwater,
they can be washed out only at the rate of groundwater turnover,
which may take decades or even centuries.

The time lags that come from slowly changing stocks
can cause problems in systems,
but they also can be sources of stability.
Soil that has accumulated over centuries
rarely erodes all at once.
A population that has learned many skills
doesn't forget them immediately.
You can pump groundwater faster than the rate it recharges
for a long time
before the aquifer is drawn down far enough to be damaged.
The time lags imposed by stocks
allow room to maneuver, to experiment,
and to revise policies that aren't working.

If you have a sense of the rates of change of stocks,
you don't expect things to happen faster than they can happen.
You don't give up too soon.
You can use the opportunities presented by a system's momentum
to guide it toward a good outcome—
much as a judo expert uses the momentum of an opponent
to achieve his or her own goals.

There is one more important principle
about the role of stocks in systems,
a principle that will lead us directly
to the concept of feedback.
The presence of stocks
allows inflows and outflows to be independent of each other
and temporarily out of balance with each other.

Stocks allow inflows and outflows to be decoupled
and to be independent
and temporarily out of balance with each other.

It would be hard to run an oil company
if gasoline had to be produced at the refinery
at exactly the rate the cars were burning it.
It isn't feasible to harvest a forest
at the precise rate at which the trees are growing.
Gasoline in storage tanks and wood in the forest
are both stocks that permit life to proceed
with some certainty, continuity, and predictability,
even though flows vary in the short term.

Human beings have invented hundreds
of stock-maintaining mechanisms
to make inflows and outflows independent and stable.
Reservoirs enable residents and farmers downriver
to live without constantly adjusting their lives and work
to a river's varying flow,
especially its droughts and floods.
Banks enable you temporarily to earn money
at a rate different from how you spend.
Inventories of products along a chain
from distributors to wholesalers to retailers
allow production to proceed smoothly
although customer demand varies,
and allow customer demand to be filled
even though production rates vary.

Most individual and institutional decisions
are designed to regulate the levels in stocks.
If inventories rise too high,
then prices are cut or advertising budgets are increased,
so that sales will go up and inventories will fall.
If the stock of food in your kitchen gets low,
you go to the store.
As the stock of growing grain rises or fails to rise in the fields,
farmers decide whether to apply water or pesticide,
grain companies decide how many barges to book for the harvest,
speculators bid on future values of the harvest,
cattle growers build up or cut down their herds.
Water levels in reservoirs cause all sorts of corrective actions
if they rise too high or fall too low.
The same can be said for the stock of money in your wallet,
the oil reserves owned by an oil company,
the pile of woodchips feeding a paper mill,
and the concentration of pollutants in a lake.

People monitor stocks constantly
and make decisions and take actions
designed to raise or lower stocks
or to keep them within acceptable ranges.
Those decisions add up to the ebbs and flows,
successes and problems, of all sorts of systems.
Systems thinkers see the world as a collection of stocks
along with the mechanisms for regulating the levels in the stocks
by manipulating flows.

That means system thinkers see the world
as a collection of "*feedback processes.*"

How the System Runs Itself—Feedback

Systems of information-feedback control
are fundamental to all life and human endeavor,
from the slow pace of biological evolution
to the launching of the latest space satellite. . . .
Everything we do as individuals,
as an industry, or as a society
is done in the context of an information-feedback system.
—Jay W. Forrester[3]

When a stock grows by leaps and bounds
or declines swiftly
or is held within a certain range
no matter what else is going on around it,
it is likely that there is a control mechanism at work.
In other words,
if you see a behavior that persists over time,
there is likely a mechanism creating that consistent behavior.
That mechanism operates through a *feedback loop*.
It is the consistent behavior pattern over a long period of time
that is the first hint of the existence of a feedback loop.

A feedback loop is formed
when changes in a stock affect the flows
into or out of that same stock.
A feedback loop can be quite simple and direct.
Think of an interest-bearing savings account in a bank.
The total amount of money in the account (the stock)
affects how much money comes into the account as interest.
That is because the bank has a rule
that the account earns a certain percent interest each year.
The total dollars of interest paid into the account each year
(the flow in)
is not a fixed amount,
but varies with the size of the total in the account.

You experience another fairly direct kind of feedback loop
when you get your bank statement
for your checking account each month.
As your level of available cash in the checking account
(a stock) goes down,
you may decide to work more hours and earn more money.
The money entering your bank account is a flow
that you can adjust
in order to increase your stock of cash
to a more desirable level.
If your bank account then grows very large,
you may feel free to work less (decreasing the inflow).
This kind of feedback loop
is keeping your level of cash available
within a range that is acceptable to you.
You can see that adjusting your earnings
is not the only feedback loop
that works on your stock of cash.
You also may be able to adjust the outflow of money
from your account, for example.
You can imagine an outflow-adjusting feedback loop for spending.

Feedback loops can cause stocks
to maintain their level within a range
or grow or decline.
In any case,
the flows into or out of the stock are adjusted
because of changes in the size of the stock itself.
Whoever or whatever is monitoring the stock's level
begins a corrective process,
adjusting rates of inflow or outflow (or both)
and so changing the stock's level.
The stock level feeds back through a chain of signals and actions
to control itself.

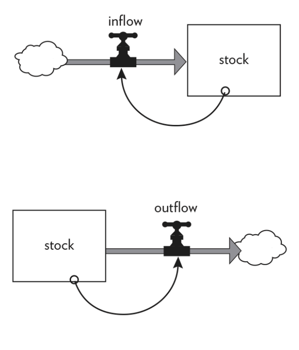

Figure 8. How to read a stock-and-flow diagram
with feedback loops.
Each diagram distinguishes the stock,
the flow that changes the stock,
and the information link
(shown as a thin, curved arrow) that directs the action.
It emphasizes that action or change
always proceeds through adjusting flows.

A feedback loop is a closed chain of causal connections
from a stock,
through a set of decisions or rules
or physical laws or actions
that are dependent on the level of the stock,
and back again through a flow to change the stock.

Not all systems have feedback loops.
Some systems are relatively simple
open-ended chains of stocks and flows.
The chain may be affected by outside factors,
but the levels of the chain's stocks don't affect its flows.
However, those systems that contain feedback loops are common
and may be quite elegant or rather surprising,
as we shall see.

Stabilizing Loops—Balancing Feedback

One common kind of feedback loop stabilizes the stock level,
as in the checking-account example.
The stock level may not remain completely fixed,
but it does stay within an acceptable range.
What follows are some more stabilizing feedback loops
that may be familiar to you.
These examples start to detail some of the steps
within a feedback loop.

If you're a coffee drinker,
when you feel your energy level run low,
you may grab a cup of hot black stuff to perk you up again.
You, as the coffee drinker,
hold in your mind a desired stock level
(energy for work).
The purpose of this caffeine-delivery system
is to keep your actual stock level
near or at your desired level.
(You may have other purposes for drinking coffee as well:
enjoying the flavor or engaging in a social activity.)
It is the gap, the discrepancy,
between your actual and desired levels of energy for work
that drives your decisions
to adjust your daily caffeine intake.

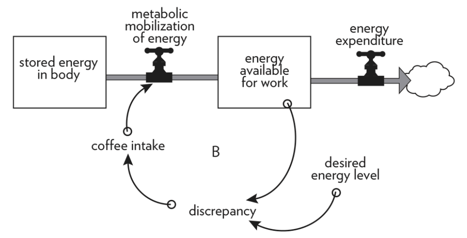

Figure 9. Energy level of a coffee drinker.

Notice that the labels in Figure 9,
like all the diagram labels in this book,
are direction-free.
The label says "*stored energy in body*"
not "*low energy level,*"
"*coffee intake*" not "*more coffee.*"
That's because feedback loops often can operate
in two directions.
In this case, the feedback loop can correct an oversupply
as well as an undersupply.
If you drink too much coffee
and find yourself bouncing around with extra energy,
you'll lay off the caffeine for a while.
High energy creates a discrepancy that says "*too much,*"
which then causes you to reduce your coffee intake
until your energy level settles down.
The diagram is intended to show
that the loop works to drive the stock of energy
in either direction.

I could have shown the inflow of energy coming from a cloud,
but instead I made the system diagram slightly more complicated.
Remember—all system diagrams
are simplifications of the real world.
We each choose how much complexity to look at.
In this example, I drew another stock—
the stored energy in the body
that can be activated by the caffeine.
I did that to indicate
that there is more to the system than one simple loop.
As every coffee drinker knows,
caffeine is only a short-term stimulant.
It lets you run your motor faster,
but it doesn't refill your personal fuel tank.
Eventually the caffeine high wears off,
leaving the body more energy-deficient than it was before.
That drop could reactivate the feedback loop
and generate another trip to the coffee pot.
(See the discussion of addiction later in this book.)
Or it could activate some longer-term
and healthier feedback responses:
Eat some food, take a walk, get some sleep.

This kind of stabilizing, goal-seeking, regulating loop
is called a *balancing feedback loop*,
so I put a B inside the loop in the diagram.
Balancing feedback loops are goal-seeking or stability-seeking.
Each tries to keep a stock at a given value
or within a range of values.
A balancing feedback loop opposes
whatever direction of change is imposed on the system.
If you push a stock too far up,
a balancing loop will try to pull it back down.
If you shove it too far down,
a balancing loop will try to bring it back up.

Here's another balancing feedback loop that involves coffee,
but one that works through physical law
rather than human decision.
A hot cup of coffee will gradually cool down
to room temperature.
Its rate of cooling depends on the difference
between the temperature of the coffee
and the temperature of the room.
The greater the difference,
the faster the coffee will cool.
The loop works the other way too—
if you make iced coffee on a hot day,
it will warm up until it has the same temperature as the room.
The function of this system
is to bring the discrepancy
between coffee's temperature and room's temperature to zero,
no matter what the direction of the discrepancy.

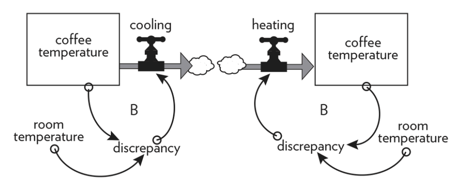

Figure 10. A cup of coffee cooling (left) or warming (right).

Starting with coffee at different temperatures,
from just below boiling to just above freezing,
the graphs in Figure 11 show what will happen
to the temperature over time
(if you don't drink the coffee).
You can see here the "*homing*" behavior
of a balancing feedback loop.
Whatever the initial value of the system stock
(coffee temperature in this case),
whether it is above or below the "*goal*" (room temperature),
the feedback loop brings it toward the goal.
The change is faster at first, and then slower,
as the discrepancy between the stock and the goal decreases.

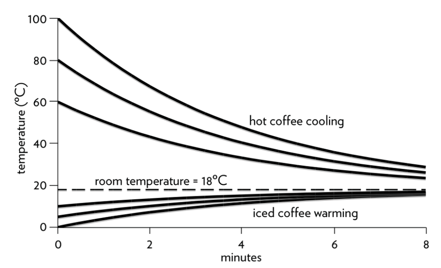

Figure 11. Coffee temperature as it approaches
a room temperature of 18°C.

Balancing feedback loops
are equilibrating or goal-seeking structures in systems
and are both sources of stability
and sources of resistance to change.

This behavior pattern—
gradual approach to a system-defined goal—
also can be seen when a radioactive element decays,
when a missile finds its target,
when an asset depreciates,
when a reservoir is brought up or down to its desired level,
when your body adjusts its blood-sugar concentration,
when you pull your car to a stop at a stoplight.
You can think of many more examples.
The world is full of goal-seeking feedback loops.

The presence of a feedback mechanism
doesn't necessarily mean that the mechanism works well.
The feedback mechanism may not be strong enough
to bring the stock to the desired level.
Feedbacks—the interconnections,
the information part of the system—
can fail for many reasons.
Information can arrive too late or at the wrong place.
It can be unclear or incomplete or hard to interpret.
The action it triggers may be too weak or delayed
or resource-constrained or simply ineffective.
The goal of the feedback loop
may never be reached by the actual stock.
But in the simple example of a cup of coffee,
the drink eventually will reach room temperature.

Runaway Loops—Reinforcing Feedback

I'd need rest to refresh my brain,
and to get rest it's necessary to travel,
and to travel one must have money,
and in order to get money you have to work. . . .
I am in a vicious circle . . .
from which it is impossible to escape.
—Honoré Balzac,[4] *19th century novelist and playwright*

Here we meet a very important feature.
It would seem as if this were circular reasoning;
profits fell because investment fell,
and investment fell because profits fell.
—Jan Tinbergen,[5] *economist*

The second kind of feedback loop
is amplifying, reinforcing, self-multiplying, snowballing—
a vicious or virtuous circle
that can cause healthy growth or runaway destruction.
It is called a *reinforcing feedback loop*,
and will be noted with an R in the diagrams.
It generates more input to a stock
the more that is already there
(and less input the less that is already there).
A reinforcing feedback loop enhances
whatever direction of change is imposed on it.

For example:
– When we were kids, the more my brother pushed me,
the more I pushed him back,
so the more he pushed me back,
so the more I pushed him back.
– The more prices go up,
the more wages have to go up
if people are to maintain their standards of living.
The more wages go up,
the more prices have to go up to maintain profits.
This means that wages have to go up again,
so prices go up again.
– The more rabbits there are,
the more rabbit parents there are to make baby rabbits.
The more baby rabbits there are,
the more grow up to become rabbit parents,
to have even more baby rabbits.
– The more soil is eroded from the land,
the less plants are able to grow,
so the fewer roots there are to hold the soil,
so the more soil is eroded,
so less plants can grow.
– The more I practice piano,
the more pleasure I get from the sound,
and so the more I play the piano,
which gives me more practice.

Reinforcing loops are found
wherever a system element has the ability to reproduce itself
or to grow as a constant fraction of itself.
Those elements include populations and economies.
Remember the example of the interest-bearing bank account?

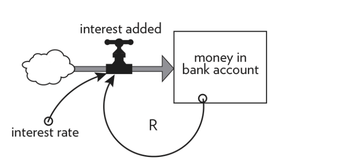

Figure 12. Interest-bearing bank account.

The more money you have in the bank,
the more interest you earn,
which is added to the money already in the bank,
where it earns even more interest.

Figure 13 shows how this reinforcing loop multiplies money,
starting with $100 in the bank,
and assuming no deposits and no withdrawals
over a period of twelve years.
The five lines show five different interest rates,
from 2 percent to 10 percent per year.

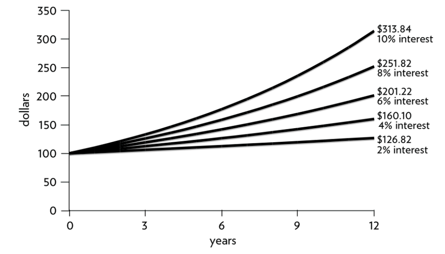

Figure 13. Growth in savings with various interest rates.

This is not simple linear growth.
It is not constant over time.
The growth of the bank account at lower interest rates
may look linear in the first few years.
But, in fact, growth goes faster and faster.
The more is there, the more is added.
This kind of growth is called *exponential*.
It's either good news or bad news,
depending on what is growing—
money in the bank,
people with HIV/AIDS,
pests in a cornfield,
a national economy,
or weapons in an arms race.

Reinforcing feedback loops are self-enhancing,
leading to exponential growth
or to runaway collapses over time.
They are found whenever a stock
has the capacity to reinforce or reproduce itself.

In Figure 14, the more machines and factories
(collectively called "*capital*") you have,
the more goods and services ("*output*") you can produce.
The more output you can produce,
the more you can invest in new machines and factories.
The more you make,
the more capacity you have to make even more.
This reinforcing feedback loop
is the central engine of growth in an economy.

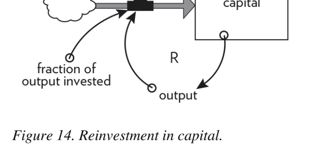

Figure 14. Reinvestment in capital.

Hint on Reinforcing Loops and Doubling Time

Because we bump into reinforcing loops so often,
it is handy to know this shortcut:
The time it takes for an exponentially growing stock
to double in size, the "*doubling time,*"
equals approximately 70 divided by the growth rate
(expressed as a percentage).

For example, if you put $100 in the bank
at 7% interest per year,
you will double your money in 10 years
($70 \div 7 = 10$).
If you get only 5% interest,
your money will take 14 years to double.

By now you may be seeing
how basic balancing and reinforcing feedback loops are to systems.
Sometimes I challenge my students
to try to think of any human decision
that occurs without a feedback loop—
that is, a decision that is made without regard
to any information about the level of the stock it influences.
Try thinking about that yourself.
The more you do,
the more you'll begin to see feedback loops everywhere.

The most common "*non-feedback*" decisions students suggest
are falling in love and committing suicide.
I'll leave it to you to decide
whether you think these are actually decisions
made with no feedback involved.

Watch out!
If you see feedback loops everywhere,
you're already in danger of becoming a systems thinker!
Instead of seeing only how A causes B,
you'll begin to wonder how B may also influence A—
and how A might reinforce or reverse itself.
When you hear in the nightly news
that the Federal Reserve Bank
has done something to control the economy,
you'll also see that the economy must have done something
to affect the Federal Reserve Bank.
When someone tells you that population growth causes poverty,
you'll ask yourself how poverty may cause population growth.

You'll be thinking not in terms of a static world,
but a dynamic one.
You'll stop looking for who's to blame;
instead you'll start asking, "*What's the system?*"
The concept of feedback opens up the idea
that a system can cause its own behavior.

So far, I have limited this discussion
to one kind of feedback loop at a time.
Of course, in real systems
feedback loops rarely come singly.
They are linked together,
often in fantastically complex patterns.
A single stock is likely to have several
reinforcing and balancing loops of differing strengths
pulling it in several directions.
A single flow may be adjusted
by the contents of three or five or twenty stocks.
It may fill one stock while it drains another
and feeds into decisions that alter yet another.
The many feedback loops in a system tug against each other,
trying to make stocks grow, die off,
or come into balance with each other.
As a result, complex systems do much more
than stay steady or explode exponentially
or approach goals smoothly—
as we shall see.

Think About This

If A causes B,
is it possible that B also causes A?